



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ASSESSMENT TOOL 3 – CODING CHALLENGE REPORT

Submitted in partial fulfillment for the Course

DSA0303 – NATURAL LANGUAGE PROCESSING

Submitted to the

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For the award of the degree of

BACHELOR OF ENGINEERING

IN

CSE (AI)

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Introduction

Natural Language Processing (NLP) enables computers to understand and process human language. In this coding challenge, Python is used to implement **Regular Expressions** and **Finite State Automata (FSA)** for validating text, recognizing patterns, and processing language data.

Question 1: Intelligent Email and Password Validator using Regular Expressions.

Objective

To validate Email Address, Password, and Mobile Number using Regular Expressions.

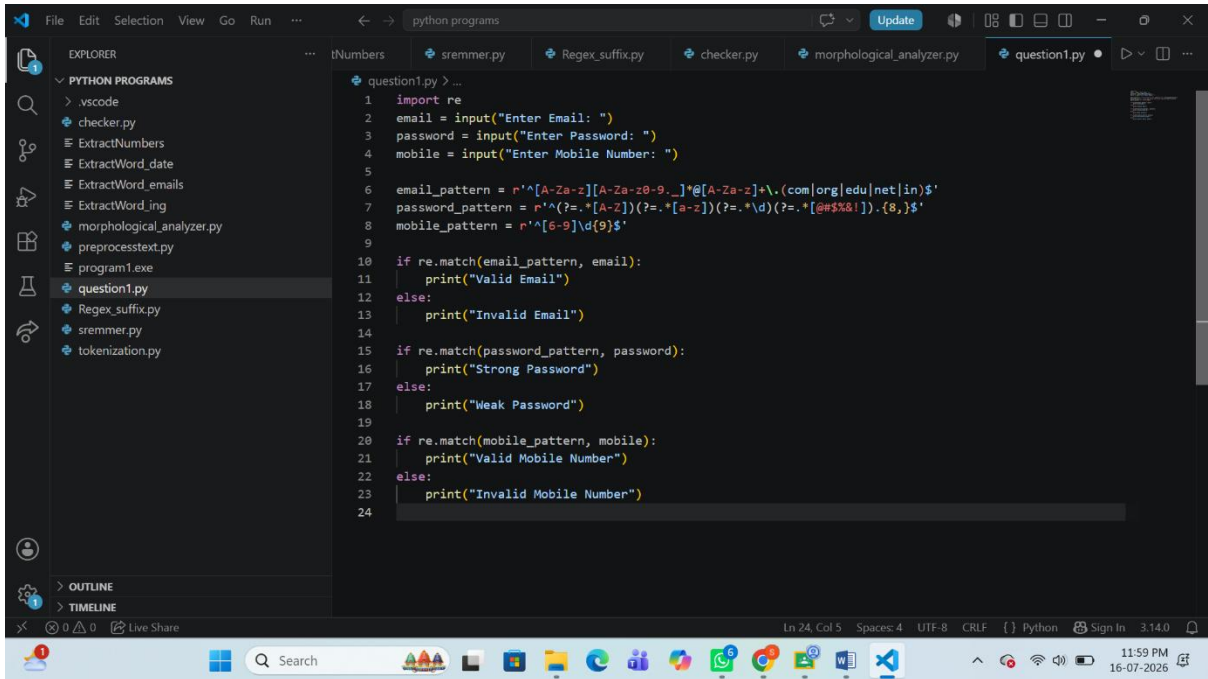
Software Used

- Python 3.x
- VS Code

Algorithm

1. Read Email, Password and Mobile Number.
2. Validate Email using Regex.
3. Validate Password using Regex.
4. Validate Mobile Number using Regex.
5. Display the result.

Simulation with output:



```
1 import re
2 email = input("Enter Email: ")
3 password = input("Enter Password: ")
4 mobile = input("Enter Mobile Number: ")
5
6 email_pattern = r'^[A-Za-z][A-Za-z0-9_]*@[A-Za-z]+\.(com|org|edu|net|in)$'
7 password_pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%&!])\{8,}$'
8 mobile_pattern = r'^[6-9]\d{9}$'
9
10 if re.match(email_pattern, email):
11     print("Valid Email")
12 else:
13     print("Invalid Email")
14
15 if re.match(password_pattern, password):
16     print("Strong Password")
17 else:
18     print("Weak Password")
19
20 if re.match(mobile_pattern, mobile):
21     print("Valid Mobile Number")
22 else:
23     print("Invalid Mobile Number")
24
```

```
PS C:\Users\Ms Tech Care\OneDrive\Desktop\python programs> python -u "c:\Users\Ms Tech Care\OneDrive\Desktop\python programs\question1.py"
Enter Email: saivarshini@gmail.com
Enter Password: 123password
Enter Mobile Number: 9866401810
Valid Email
Weak Password
Valid Mobile Number
PS C:\Users\Ms Tech Care\OneDrive\Desktop\python programs> 
```

Question 2: Finite State Automata (FSA) Simulator

Objective

To check whether the entered string ends with "**ab**".

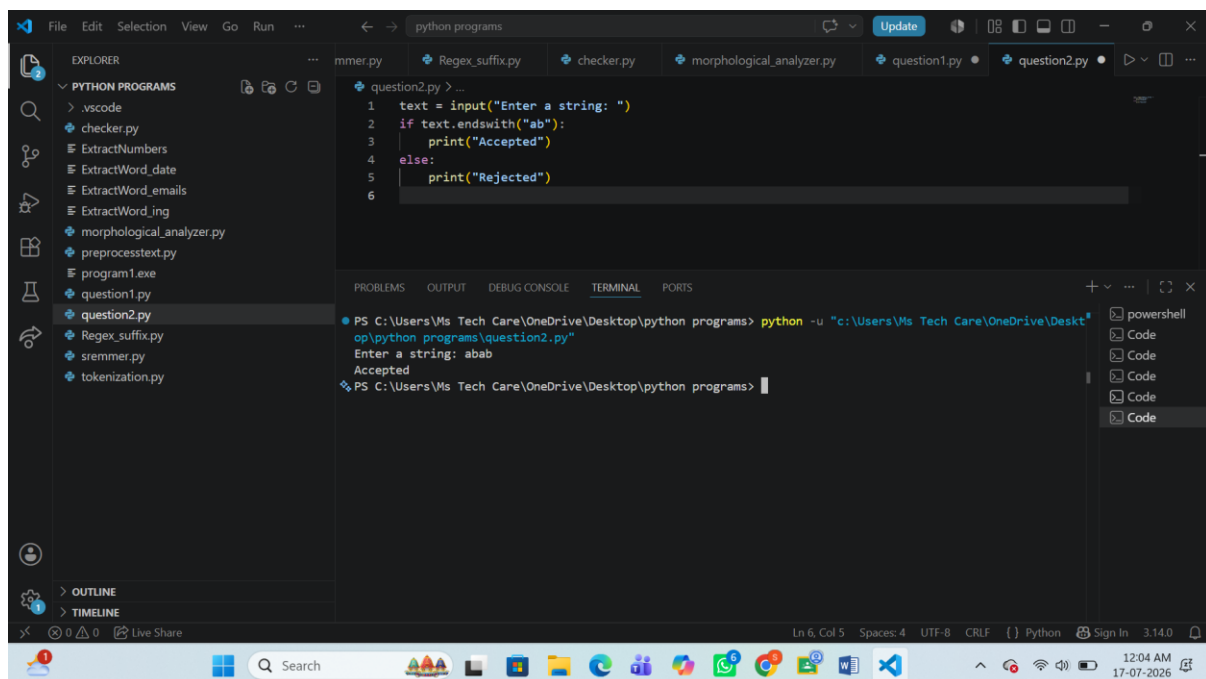
Software Used

- Python
- VS Code

Algorithm

1. Read a string.
2. Check whether the string ends with **ab**.
3. If yes, display Accepted.
4. Otherwise display Rejected.

Simulation with output:



The screenshot displays the Visual Studio Code interface. The Explorer panel on the left shows a project named 'python programs' containing several files, including 'question2.py'. The main editor window shows the code for 'question2.py':

```
1 text = input("Enter a string: ")
2 if text.endswith("ab"):
3     print("Accepted")
4 else:
5     print("Rejected")
6
```

The TERMINAL panel at the bottom shows the command prompt output:

```
PS C:\Users\Ms Tech Care\OneDrive\Desktop\python programs> python -u "c:\Users\Ms Tech Care\OneDrive\Desktop\python programs\question2.py"
Enter a string: abab
Accepted
PS C:\Users\Ms Tech Care\OneDrive\Desktop\python programs>
```

The status bar at the bottom indicates the current line and column as 'Ln 6, Col 5'.

Question 3: Smart Pattern Matching Engine using Regular Expressions.

Objective

To search Dates, Phone Numbers, Hashtags and Mentions from a paragraph.

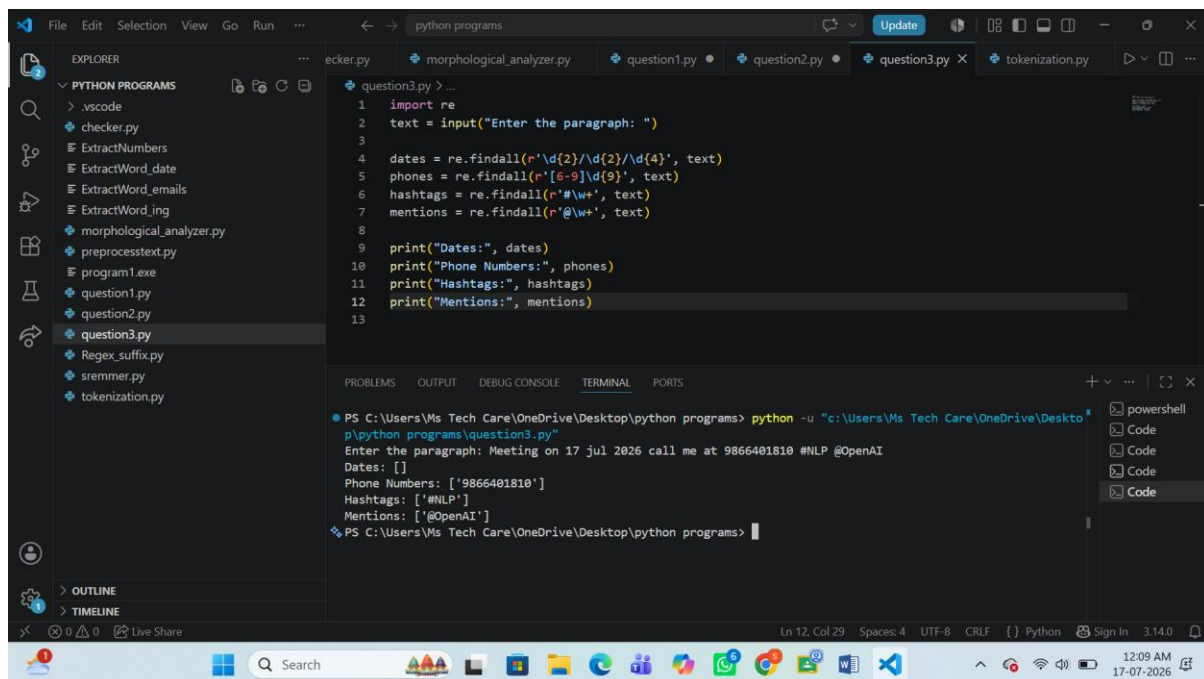
Software Used

- Python
- VS Code

Algorithm

1. Read a paragraph.
2. Search Date.
3. Search Phone Number.
4. Search Hashtag.
5. Search Mention.
6. Display the result.

Simulation with output:



The screenshot displays the Visual Studio Code interface with a Python script named `question3.py` open. The script uses regular expressions to extract dates, phone numbers, hashtags, and mentions from a user-provided paragraph. The terminal output shows the results of running the script on the sample text: "Meeting on 17 jul 2026 call me at 9866401810 #NLP @OpenAI".

```
1 import re
2 text = input("Enter the paragraph: ")
3
4 dates = re.findall(r'\d{2}/\d{2}/\d{4}', text)
5 phones = re.findall(r'[6-9]\d{9}', text)
6 hashtags = re.findall(r'#\w+', text)
7 mentions = re.findall(r'@\w+', text)
8
9 print("Dates:", dates)
10 print("Phone Numbers:", phones)
11 print("Hashtags:", hashtags)
12 print("Mentions:", mentions)
13
```

Terminal Output:

```
PS C:\Users\Ms Tech Care\OneDrive\Desktop\python programs> python -u "c:\Users\Ms Tech Care\OneDrive\Desktop\python programs\question3.py"
p:\python programs\question3.py
Enter the paragraph: Meeting on 17 jul 2026 call me at 9866401810 #NLP @OpenAI
Dates: []
Phone Numbers: ['9866401810']
Hashtags: ['#NLP']
Mentions: ['@OpenAI']
PS C:\Users\Ms Tech Care\OneDrive\Desktop\python programs>
```

